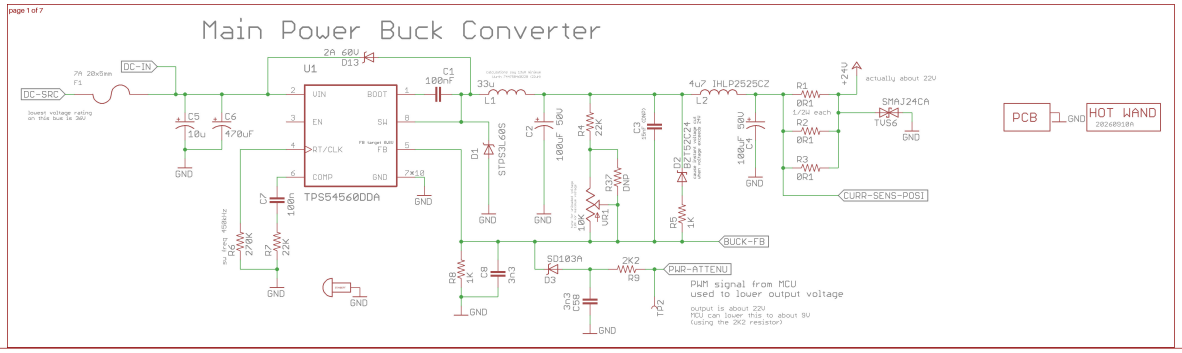
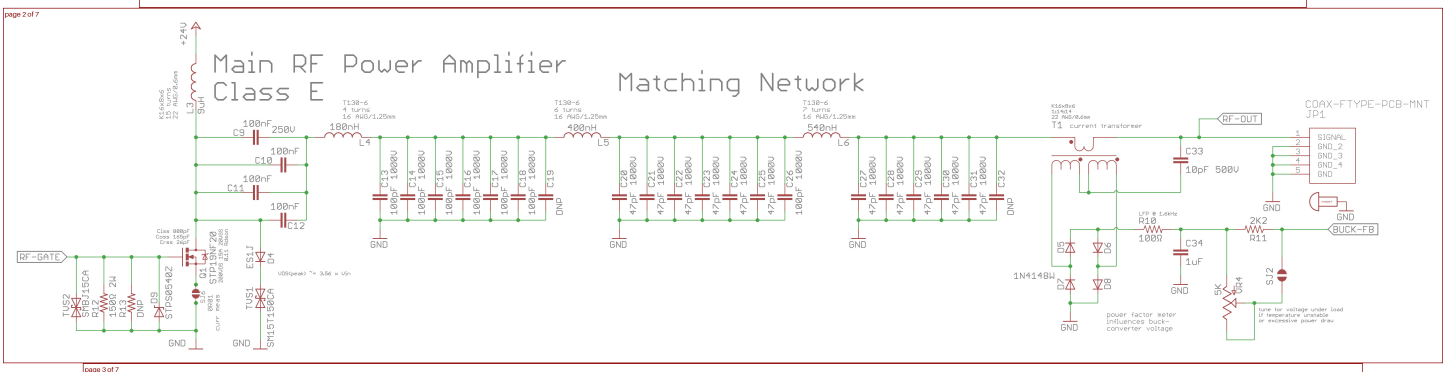
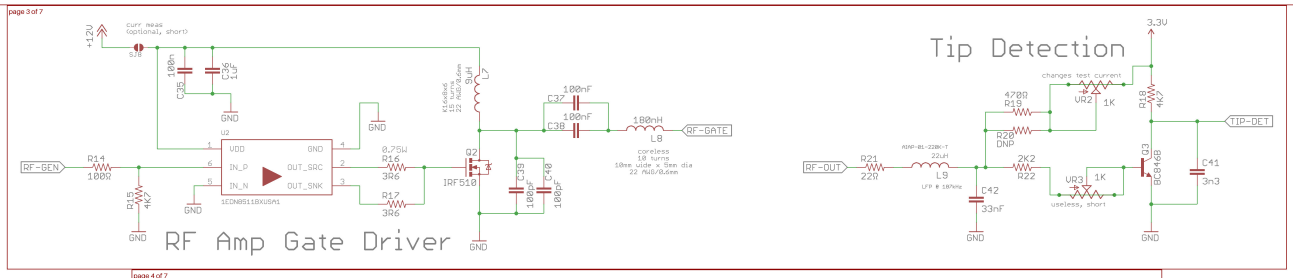
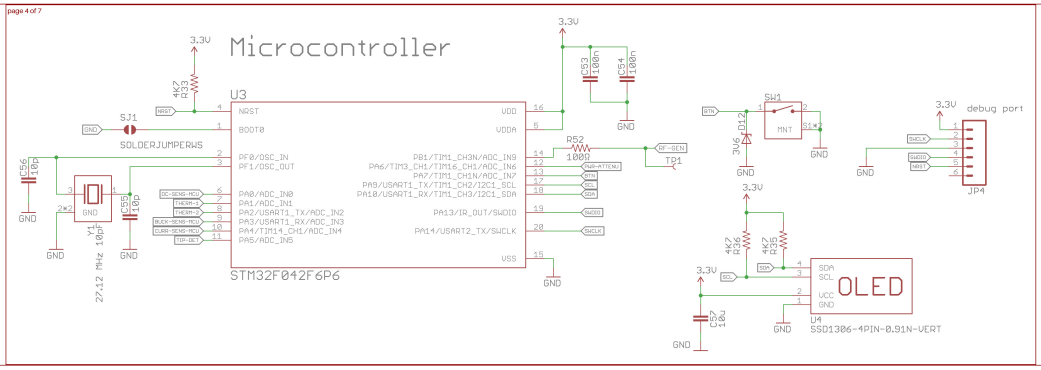
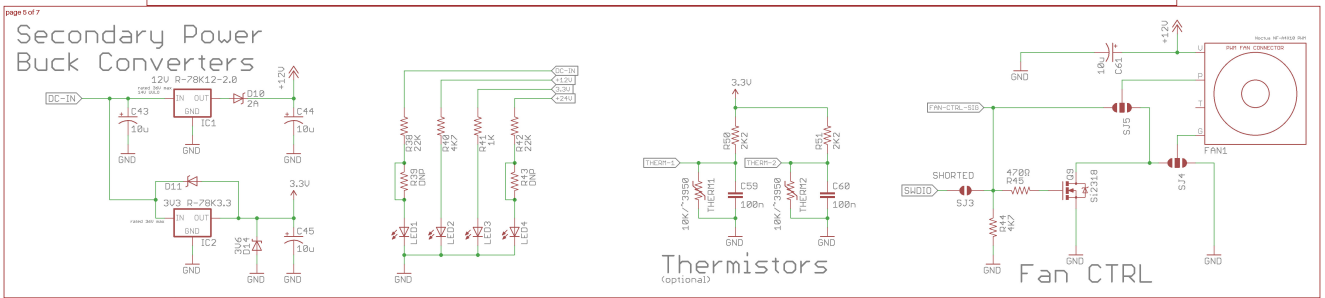
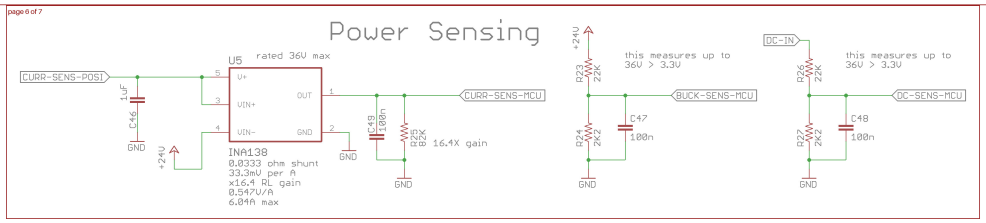

page 1 of 7
Main Power Buck Converter
DC-IN
DC-SRC
U1
TPS54560DDA
D13
D1
STPS3L60S
L1
33U
L2
4u7 IHLP2525CZ
D12
BZT52C22
SMAJ24CA
TVS6
PCB
GND
HOT WAND
BUCK-FB
PWR-ATTENU
PWM signal from MCU
used to lower output voltage
output is about 22V
MCU can lower this to about 9V
(using the 2K2 resistor)
CURR-SENS-POSI

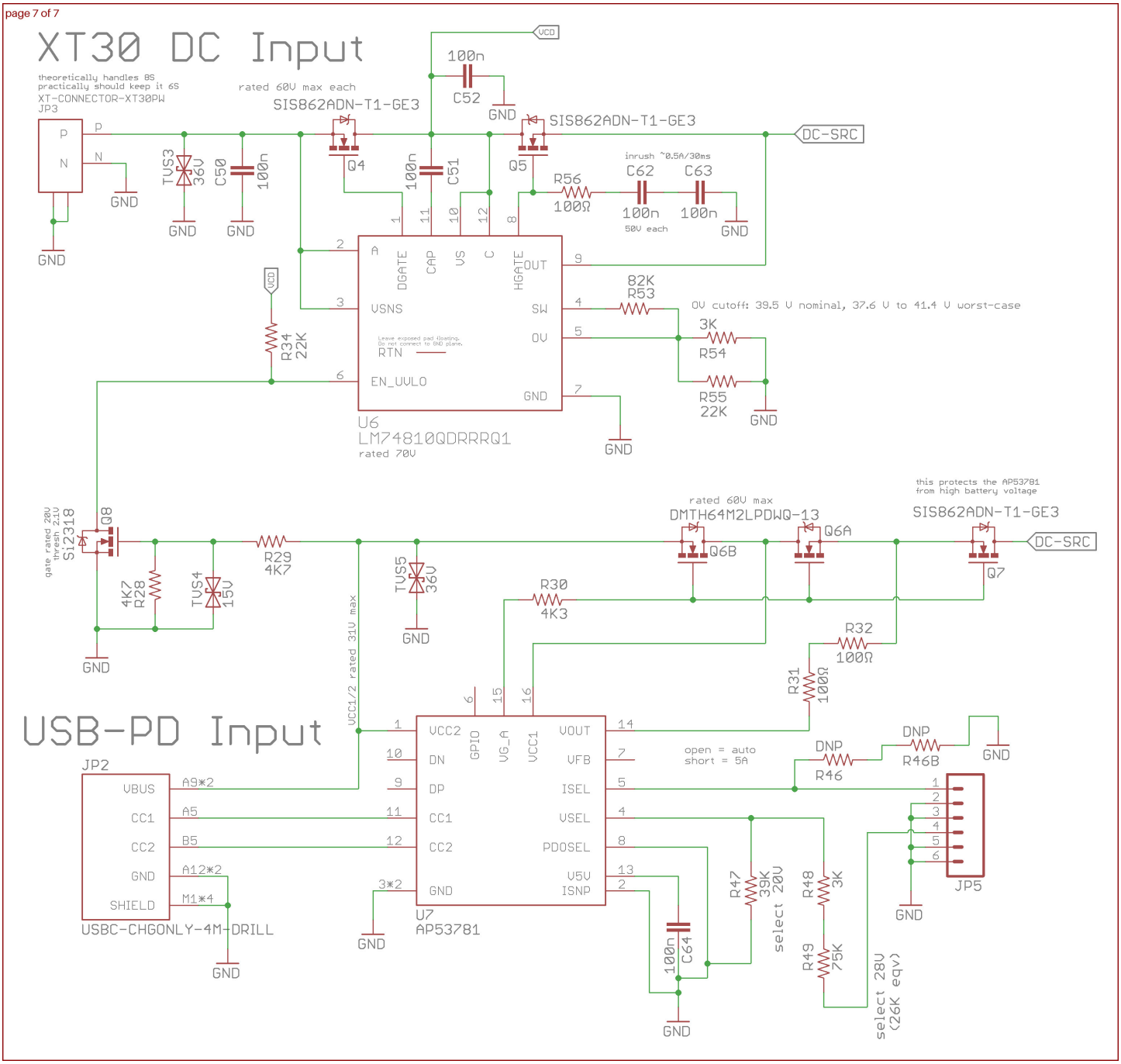

page 2 of 7
Main RF Power Amplifier
Class E
Matching Network
RF-GATE
RF-OUT
COAX-FTYPE-PCB-MNT
JP1
SIGNAL
GND_2
GND_3
GND_4
GND
T1 current transformer
BUCK-FB
power factor meter
influences buck-
converter voltage
tune for voltage under load
if temperature unstable
or excessive power draw


page 3 of 7
RF Amp Gate Driver
Tip Detection
RF-GEN
RF-OUT
RF-GATE
TIP-DET
changes test current
useless, short


page 4 of 7
Microcontroller
STM32F042F6P6
SOLDERJUMPERWS
debug port
JP4
OLED
SSD1306-4PIN-0.91N-VERT


page 5 of 7
Secondary Power
Buck Converters
12V R-78K12-2.0
3V3 R-78K3.3
Thermistors
(optional)
Fan CTRL
FAN1
SHORTED


page 6 of 7
Power Sensing
rated 36V max
INA138
0.0333 ohm shunt
33.3mV per A
x16.4 RL gain
0.547V/A
6.04A max
16.4X gain
this measures up to
36V > 3.3V
CURR-SENS-MCU
BUCK-SENS-MCU
DC-SENS-MCU


XT30 DC Input
theoretically handles 8S
practically should keep it 6S
XT-CONNECTOR-XT30PW
JP3
P
N
GND
TVS3
36V
C50
100n
rated 60V max each
SIS862ADN-T1-GE3
Q4
VCD
100n
C52
C51
Q5
DC-SRC
inrush ~0.5A/30ms
C62
C63
100n
100n
50V each
R56
100Ω
U6
LM74810QDRRRQ1
rated 70V
DGATE
CAP
VS
C
HGATE
OUT
VSNS
SW
OV
RTN
EN_UVLO
GND
R34
22K
82K
R53
3K
R54
R55
22K
OV cutoff: 39.5 V nominal, 37.6 V to 41.4 V worst-case
Leave exposed pad floating.
Do not connect to GND plane.
this protects the AP53781
from high battery voltage
rated 60V max
DMTH64M2LPDWQ-13
Q6A
Q6B
SIS862ADN-T1-GE3
Q7
Q8
gate rated 20V
thresh 2.1V
SI2318
4K7
R28
TVS4
15V
R29
4K7
TVS5
36V
R30
4K3
R32
100Ω
R31
100Ω
VCC1/2 rated 31V max
USB-PD Input
JP2
VBUS
CC1
CC2
GND
SHIELD
A9*2
A5
B5
A12*2
M1*4
USBC-CHGONLY-4M-DRILL
U7
AP53781
VCC2
DN
DP
CC1
CC2
GND
GPIO
VG_A
VCC1
VOUT
VFB
ISEL
VSEL
PDOSEL
V5V
ISNP
3*2
open = auto
short = 5A
DNP
R46
DNP
R46B
JP5
R47
39K
select 20V
R48
3K
R49
75K
select 28V
(26K eqv)
100n
C64